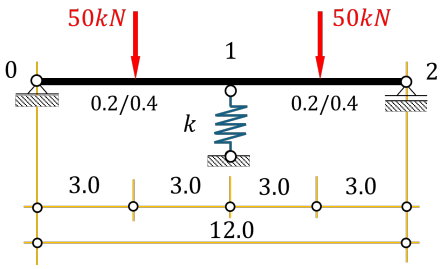
Студијски модул: АИ, ХиВИО, С 13.05.2026.

1: За носач на скици када је у чвору "1", еластичан, односно крут ослонац, срачунати и нацртати:

- статичку и кинематичку класификацију,
- реакције у ослонцима и силе у пресецима,
- вертикално померање чвора на половини распона "0-1",
- за колико је већи угиб у чвору "1".

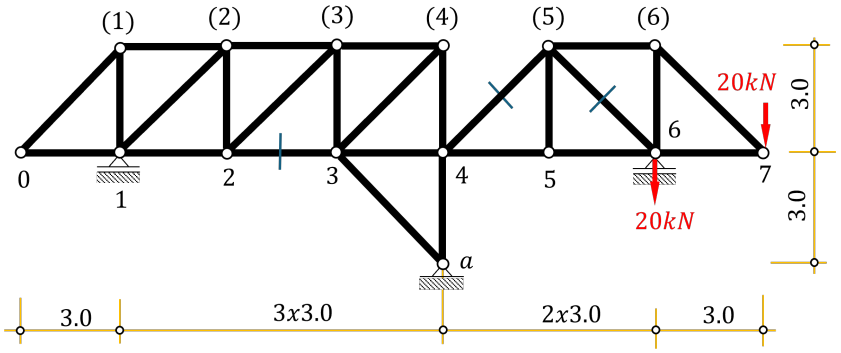
Подаци: $E = 3 \cdot 10^7 \frac{kN}{m^2}$, $k = \frac{286}{EI}$, занемарити N и T – силе на деформацију.



2: За носач приказан на скици, срачунати и нацртати:

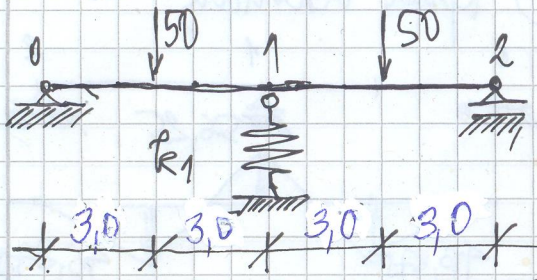
- силе у штаповима користећи аналитичке изразе,
- вертикално померање чвора "5".

Подаци: $E = 3 \cdot 10^7 \frac{kN}{m^2}$, штапови $b/h = 20/20cm$,



Предметни наставник:
проф. др Илија М. МИЛИЧИЋ, дипл.инж.грађ.

(10) ЗАДАТАК

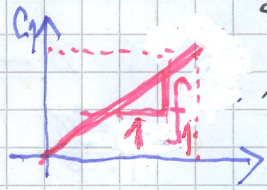


$k_1 = \frac{286}{EI}$ $E = 3 \cdot 10^7$ $I = \frac{0.2 \cdot 0.4^3}{12}$

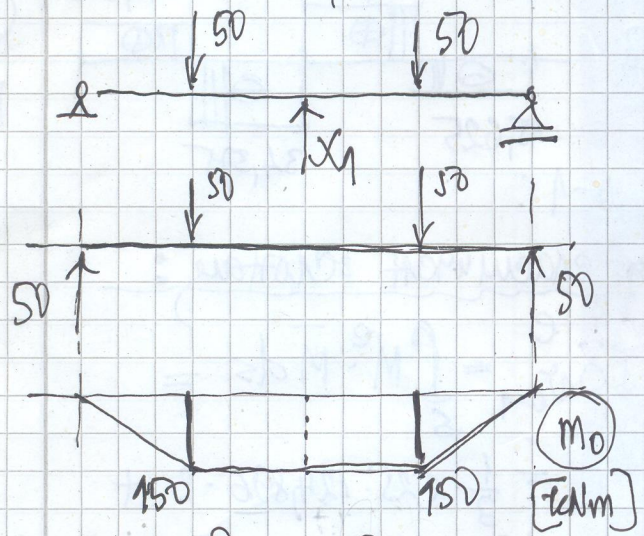
$EI = 32000$

$k_1 = \frac{286}{32000} = 8{,}937 \cdot 10^{-3} \frac{kN}{m}$

$f_1 = \frac{1}{k} = 111{,}888 \frac{m}{kN}$



* ОСНОВНИ СИСТЕМ

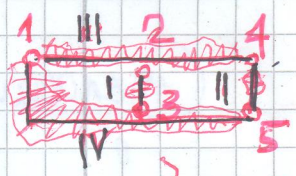


$\delta_{11} X_1 + \delta_{10} = -f_1 \cdot X_1$

$(36 + 111{,}888) X_1 - 2475 = 0$

$X_1 = 16{,}736$ kN

* ПРВИ НАЧИН *



$z_0 + z_u + z_k + z_s \overset{>}{=} 2k$

$4 + 0 + 1 + 2 > 2 \cdot 3$

$7 > 6$

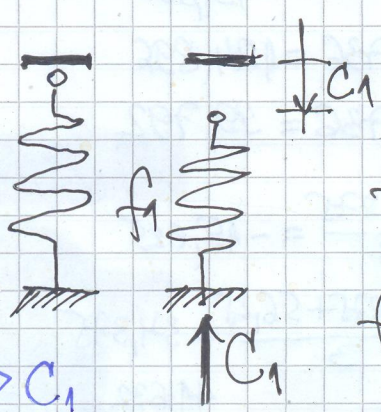
$n = 7 - 6 = 1$ X С.Н.С

$2 \cdot z_z > 3p - 3$

$2 \cdot 5 > 3 \cdot 4 - 3$

$10 > 9$

У.К.В.С

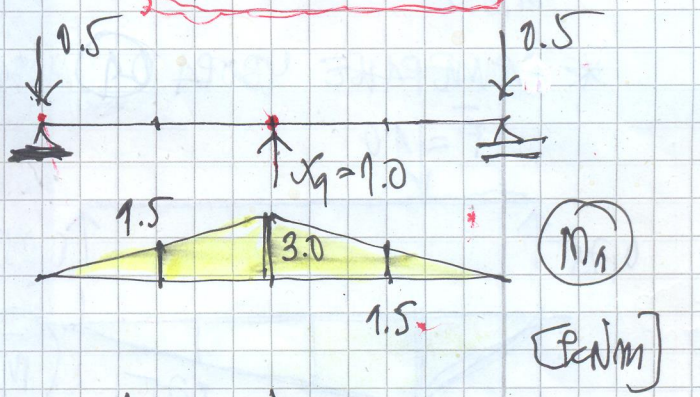


$f_1 = \frac{C_1}{C_1} \left(\frac{m}{kN} \right)$

$f_1 = \frac{1}{k} \left(\frac{m}{kN} \right)$

* ЕЛАСТИЧАН ОСЛОНАЦ

$\vec{F}_{ex} = -k \cdot x \cdot \vec{i}$ — еластична сила.



$\delta_{11} = 2 \cdot \left(\frac{1}{3} \cdot 3 \cdot 3 \cdot 6 \right) = 36$

$\delta_{10} = 2 \cdot \left[-\frac{1}{3} 150 \cdot 1{,}5 \cdot 3 - \frac{1}{2} \cdot 150 (1{,}5 + 5) \cdot 3 \right]$

$\delta_{10} = -2475$

Померање тачке на месту обртуће је $X_1 = k \cdot x \Rightarrow x = X_1 / k$

$\frac{x}{EI} \cdot 1000 = 58{,}517$ mm

* КРУТИ ОСЛОНАЦ

$\delta_{11} \cdot X_1 + \delta_{10} = 0$

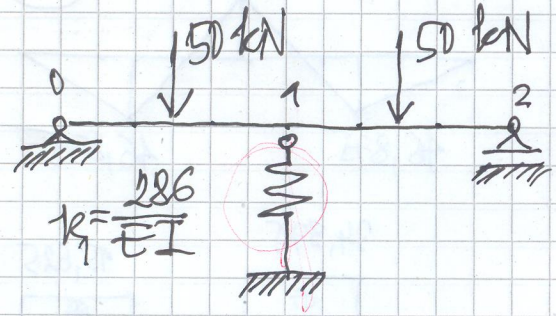
$36 \cdot X_1 - 2475 = 0$

$X_1 = 68{,}75$ kN

Односно, померање чвора 1

$\delta v_1 = 58{,}517$ mm

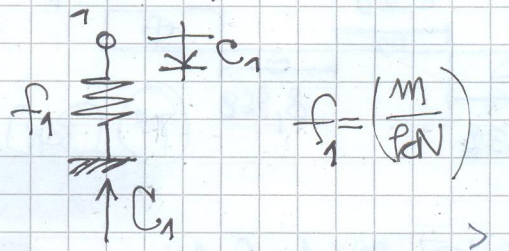
* ДРУГИ НАЧИН



$E = 3 \cdot 10^7 \quad I = \frac{0.2 \times 0.4^3}{12} = 1.067 \cdot 10^{-3}$

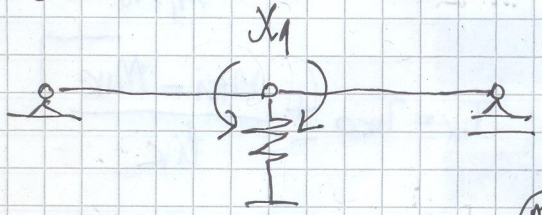
$k_1 = 8{,}937 \cdot 10^{-3} \ \frac{kN}{m}$

$f_1 = \frac{1}{k_1} = 111.888 \ \frac{m}{kN}$



$f_1 = \left(\frac{m}{kN}\right)$

* ОСНОВНА СИСТЕМА



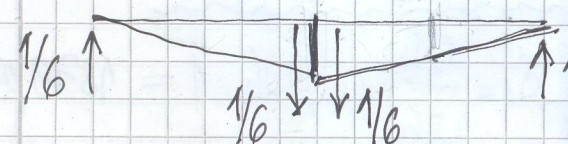
$I_0 + I_u + I_k + I_s \geq 2k$

$4 + 0 + 1 + 2 \quad 2 \cdot 3$

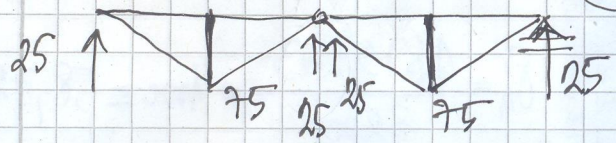
$7 > 6$

$n = 7 - 6 = 1 \times$ С.Н.С.

(M_1) [kNm]



(M_0) [kNm]



* ЕЛАСТИЧНИ ОСЛОНЦИ

$\delta_{11} = 2 \cdot \frac{1}{3} \cdot 1 \cdot 1 \cdot 6 + \frac{EI}{286} \cdot \left(\frac{1}{3}\right)^2 = 16{,}432$

$\delta_{10} = 2 \cdot \left[\frac{1}{3} \cdot 0{,}5 \cdot 75 \cdot 3\right] + 2 \cdot \left[\frac{1}{6} \cdot 75 \cdot (1 + 2 \cdot 0{,}5) \cdot 3\right] + \frac{EI}{286} \cdot \left(-\frac{1}{3}\right) \cdot 50 = 1639{,}802$

$\delta_{11} \cdot X_1 + \delta_{10} = 0$

$16{,}432 \cdot X_1 + 1639{,}802 = 0$

$X_1^e = 99{,}793$

$M = M_0 + M_1 \cdot X_1 =$

$= 75 + 0.5 \cdot 99{,}793 = 124{,}897$

$R_1^e = R_0 + R_1 \cdot X_1 =$

$= 50 - \frac{1}{3} \cdot 99{,}793 = 16{,}736$ kN

* КРУТИ ОСЛОНЦИ

$\delta_{11} = 2 \cdot \frac{1}{3} \cdot 1 \cdot 1 \cdot 6 = 4{,}0$ ✓

$\delta_{10} = 2 \cdot \left[\frac{1}{3} \cdot 0{,}5 \cdot 75 \cdot 3\right] + 2 \cdot \left[\frac{1}{6} \cdot 75 \cdot (1 + 2 \cdot 0{,}5) \cdot 3.0\right] =$

$= 225$

$\delta_{11} \cdot X_1 + \delta_{10} = 0$

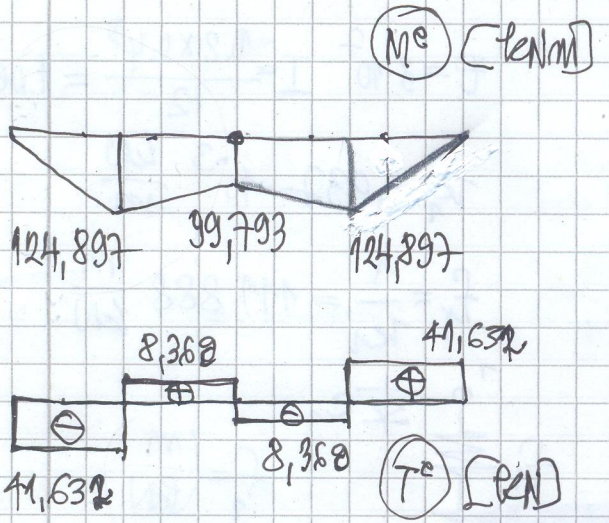
$4 \cdot x_1 + 225 = 0$

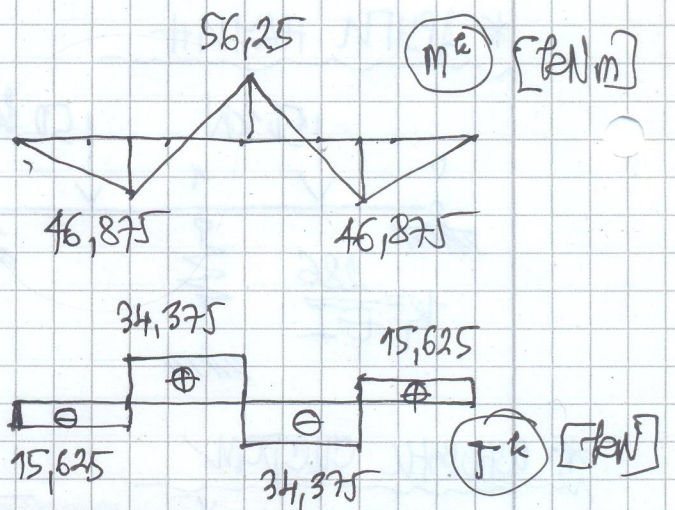
$X_1^k = -56{,}25$

$M = M_0 + M_1 \cdot X_1$

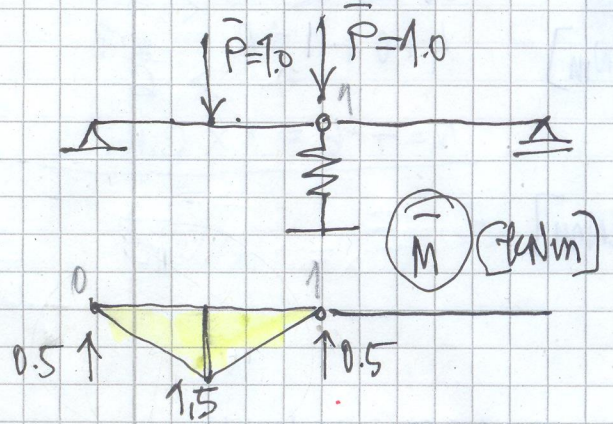
$M = 75 + 0.5 \cdot (-56{,}25) = 46{,}875$

$R_1^k = 50 - \frac{1}{3} \cdot (-56{,}25) = 68{,}75$ kN.





ПОМЕРАЊЕ МЕЂА 0-1:



$$T_{ik} = T_{ik0} \pm \frac{M_{ki} - M_{ik}}{l_{ik}}$$

✱ ПОМЕРАЊЕ ЧВОРА 1

$$EI\,\delta_1 = \frac{EI}{286} \cdot 16,796 \cdot 1 = 1879,273$$

$$\delta_1 = \frac{1879,273}{32000} \cdot 1000 = 58,727 \text{ mm}$$

1° КАДА СУ ЕЛАСТИЧНИ ОСЛОНЦИ

$$\delta_e = \frac{1}{3} \cdot 124,897 \cdot 1.5 \cdot 3 + \frac{1}{6} \cdot 1.5 \cdot [99,793 + 2 \cdot 124,897] \cdot 3 +$$
$$+ \frac{EI}{286} \cdot 16,736 \cdot 0,5 = 1385,797$$

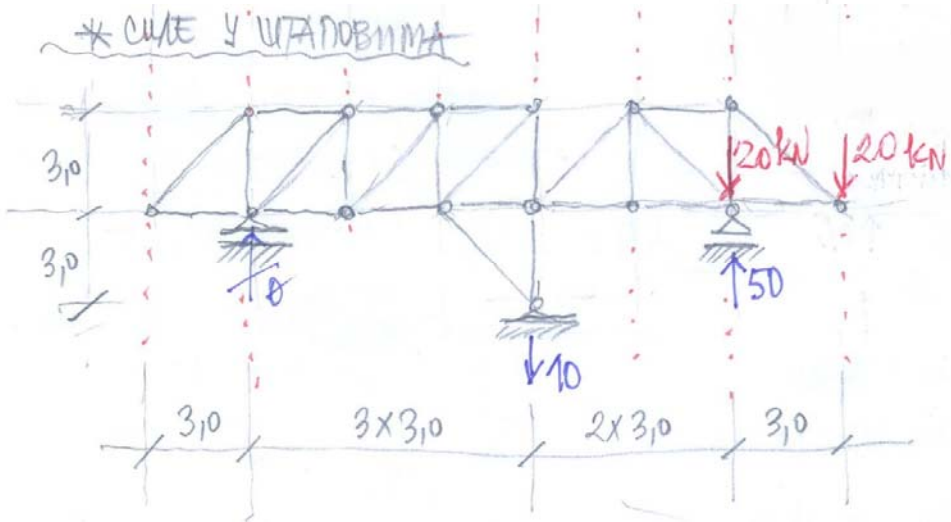
$$\delta_k = \frac{1}{3} \cdot 46,875 \cdot 1,5 \cdot 3 + \frac{1}{6} \cdot 1,5 \cdot (-56,25 + 2 \cdot 46,875) \cdot 3 = 98,438$$

ЗАКЉУЧАК: $\frac{\delta_e}{\delta_k} = \frac{1385,797}{98,438} = 14,078$ x ЈЕ ВЕЋИ КАДА СУ ЕЛАСТИЧНИ ОСЛОНЦИ!

М.Миријет

② ЗАДАТАК

* СИЛЕ У ШТАПОВИМА



$$O_{(6)} = -\frac{M_5}{h_5} = -\frac{1}{3}\cdot(-10\cdot 6) = 20\ kN$$

$$U_5 = \frac{M_{(5)}}{h_5} = \frac{1}{3}\cdot -10\cdot 3 = -10\ kN$$

$$U_6 = \frac{M_{(5)}}{h_6} = \frac{1}{3}\cdot -10\cdot 3 = -10\ kN$$

$$U_7 = \frac{M_{(6)}}{h_6} = \frac{1}{3}\cdot -10\cdot 6 = -20\ kN$$

$\alpha = 45°$ $\quad \frac{1}{\cos\alpha} = \sqrt{2}$

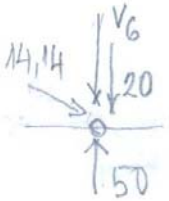
$$D_{(5)} = \left(\frac{M_4}{h_4} - \frac{M_{(5)}}{h_5}\right)\frac{1}{\cos\alpha_5} = -\frac{1}{3}\cdot -10\cdot\sqrt{2} = 14{,}14\ kN$$

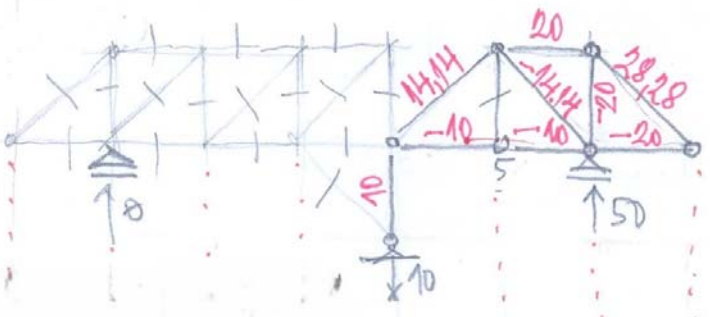
$$D_6 = \left(\frac{M_6}{h_6} - \frac{M_{(5)}}{h_5}\right)\cdot\frac{1}{\cos\alpha_6} = \frac{1}{3}\left[-10\cdot 6 + 10\cdot 3\right]\sqrt{2} = -14{,}14\ kN$$

$$D_7 = \left(\frac{M_7}{h_7} - \frac{M_{(6)}}{h_6}\right)\cdot\frac{1}{\cos\alpha_7} = -\frac{1}{3}\left[-10\cdot 6\right]\cdot\sqrt{2} = 28{,}28\ kN$$

$V_5 = 0$ (неопитерећен чвор)

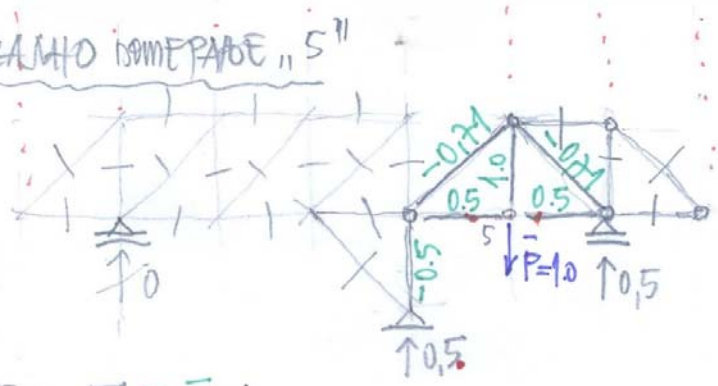
$V_6 = 50 - 20 - 14{,}14\cdot 0{,}71 = 20\ kN$





(S) (kN)

* BEPTIKALHO POMEPAHE „5"



$$EFv_5 = \sum S \cdot \bar{S} \cdot l =$$

$$= 2 \cdot (-10) \cdot 0{,}5 \cdot 3 + 10 \cdot (-0{,}5) \cdot 3 = -45$$

$$v_5 = \frac{-45}{3 \cdot 10^7 \cdot 0{,}2 \cdot 0{,}2} = -\frac{45}{1200000} \cdot 1000 = -0{,}038 \text{ mm}$$

$v_5 = -0{,}04$ mm